

Lab Num: Seven

**Topic: Create Your Own Initramfs and
Reach a Working Shell**

(No More Kernel Panic)

General Task Description:

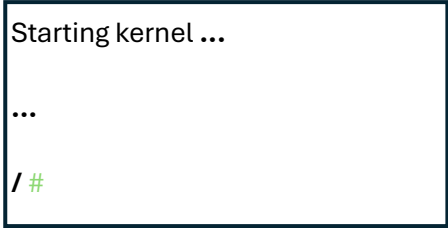
You have reached the famous line in Lab 06: [Kernel panic - not syncing: No working init found.](#)`

Now it's time to kill that panic forever.

Your mission

Build a complete, working **initramfs** (initial RAM filesystem) from scratch and make your custom kernel (from Lab 06) boot into a real shell automatically.

When you finish this lab, the moment your board powers on or QEMU starts, you must see:



```
Starting kernel ...
```

```
...
```

```
/ #
```

Exact Requirements – What You Must Deliver

1. A working initramfs archive (named rootramfs.cpio or YourName-initramfs.cpio) that contains:

- A statically-linked shell (dash) placed as /sbin/init
- Minimal directory structure so the shell can run

2. Your U-Boot bootscript must be updated so that it automatically:

- Loads your custom kernel
- Loads your .dtb
- Loads your initramfs (from SD card)

3. The system must boot to a usable shell prompt on:

- QEMU vexpress-A9

OR

- Real Raspberry Pi 3B+ (via serial console)

4. From the shell you must be able to run at least these commands successfully:

- ls
- echo "Hello from [YourName]"

Understanding Check – Questions

1. What is initramfs? Why use it instead of mounting the real rootfs directly?
2. Why cpio format for initramfs? Why not tar or zip?
3. What does **rdinit=** do? What happens if wrong path?
4. Why must init be statically linked? What if dynamic?
5. Difference: initramfs vs initrd ?
6. Where is initramfs loaded in memory? Who decompresses it?
7. How does kernel switch from initramfs to real rootfs?